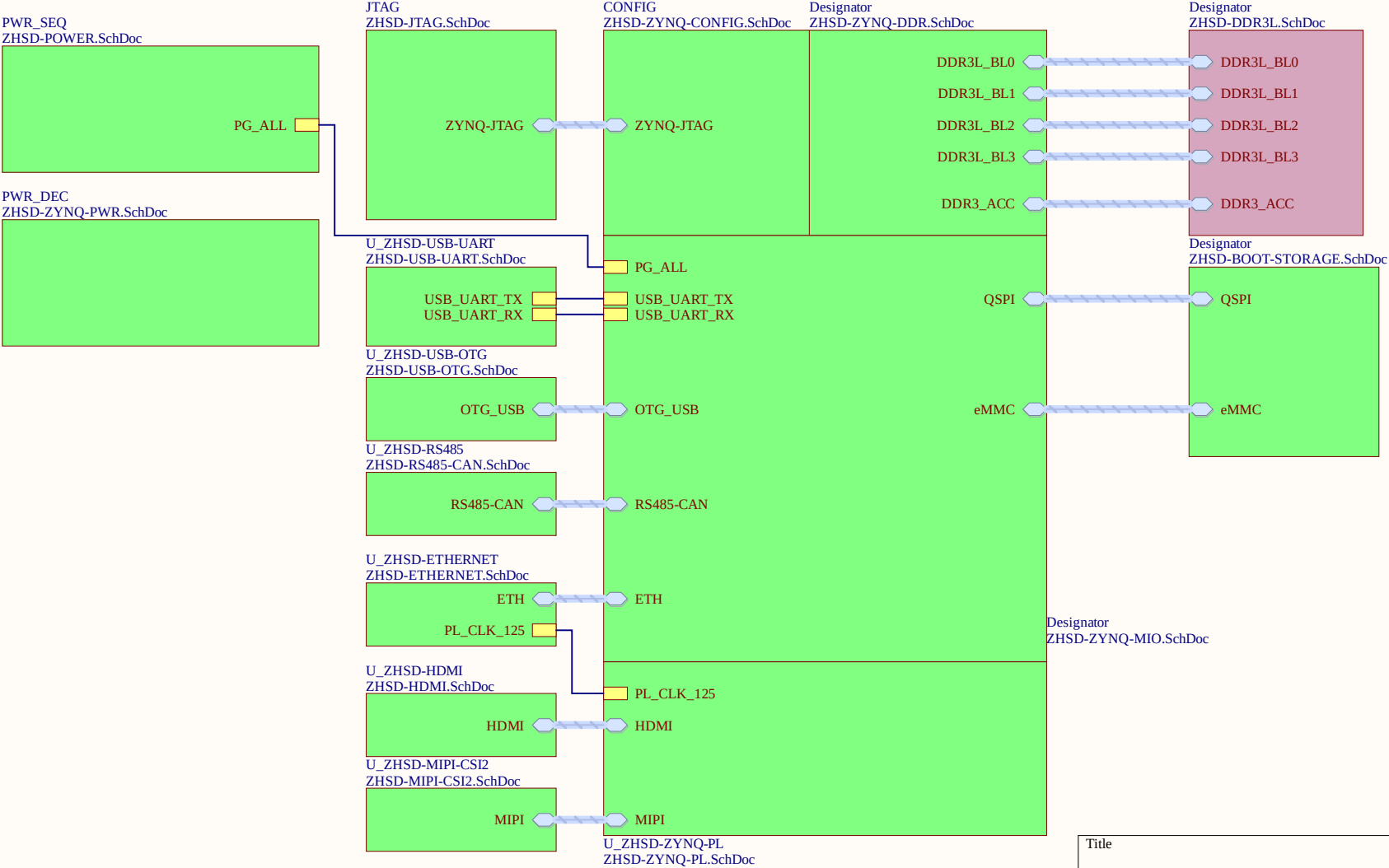


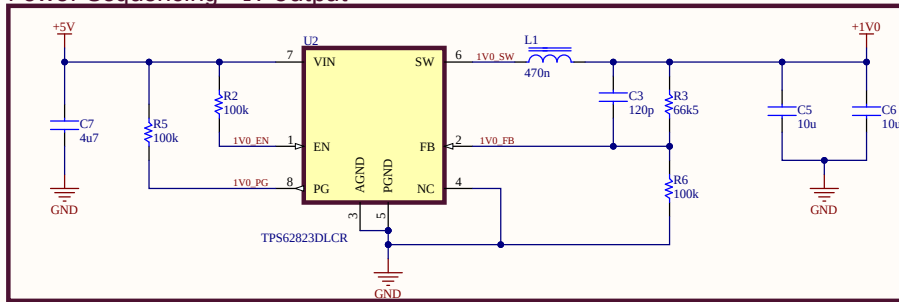
ZYNQ-7000 HIGH SPEED DESIGN



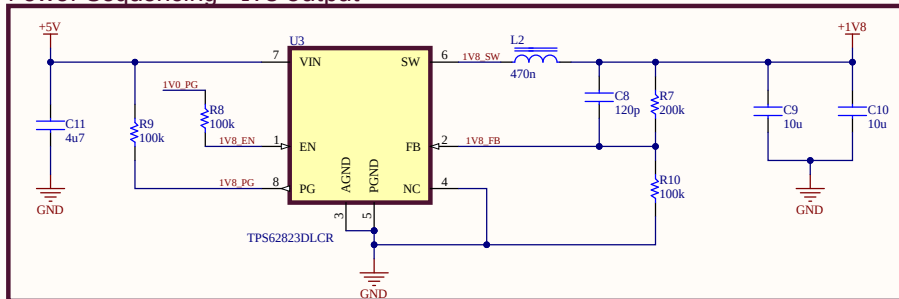
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Size	Number	Revision
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Date:	1.30.2026	Sheet of
File:	C:\Users\...\ZHSD-Block.SchDoc	Drawn By:

ZHSD POWER SYSTEM

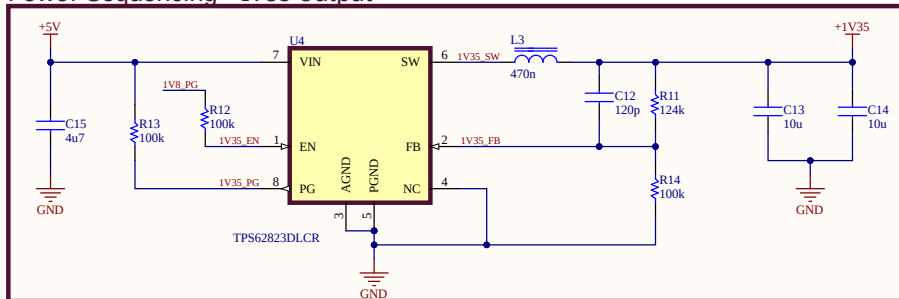
Power Sequencing - 1V Output



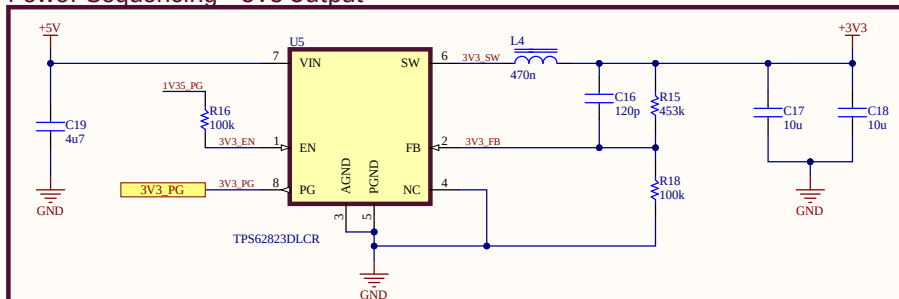
Power Sequencing - 1V8 Output



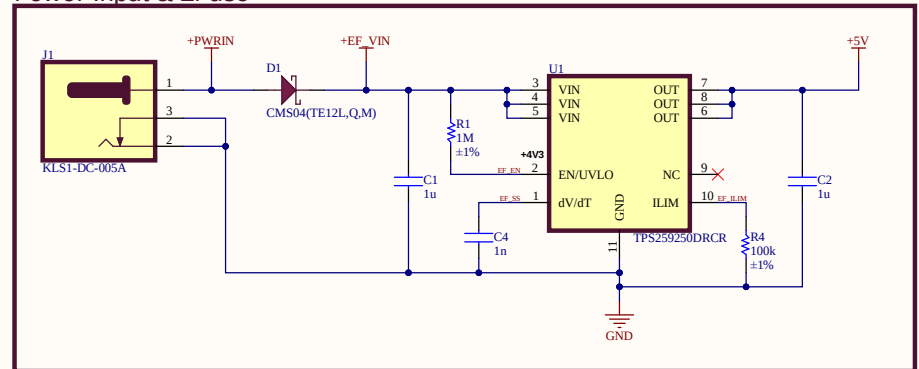
Power Sequencing - 1V35 Output



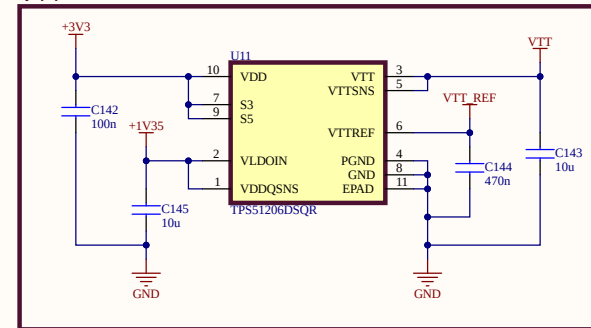
Power Sequencing - 3V3 Output



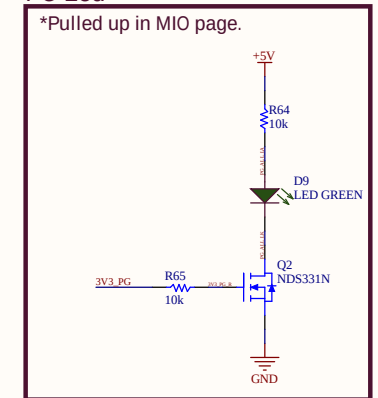
Power Input & EFuse



VTT



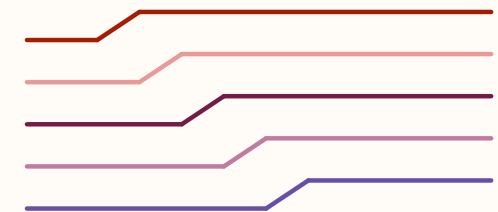
PG Led



Power Sequence

*Zynq-7000 SoC DC and AC Switching Characteristics DS187

- +1V0 VCCPINT, VCCINT, VCCBRAM
- +1V8 VCCPAUX, VCCPLL, VCCO_MIO1, VCCAUX
- +1V35 VCCO_DDR
- +3V3 VCCO_MIO0, VCCO(PL Banks)
- +VTT DDR3L Reference Voltage



Project Title		
ZYNQ-HS-DEV.PrjPcb		
Drawn By	Current Date	Modified Date
*	30.01.2026	30.01.2026
Sheet Name	Sheet Number / Sheet Total	Sheet Size
ZHSD-POWER.SchDoc	2 / 16	A4

A

3

C



D

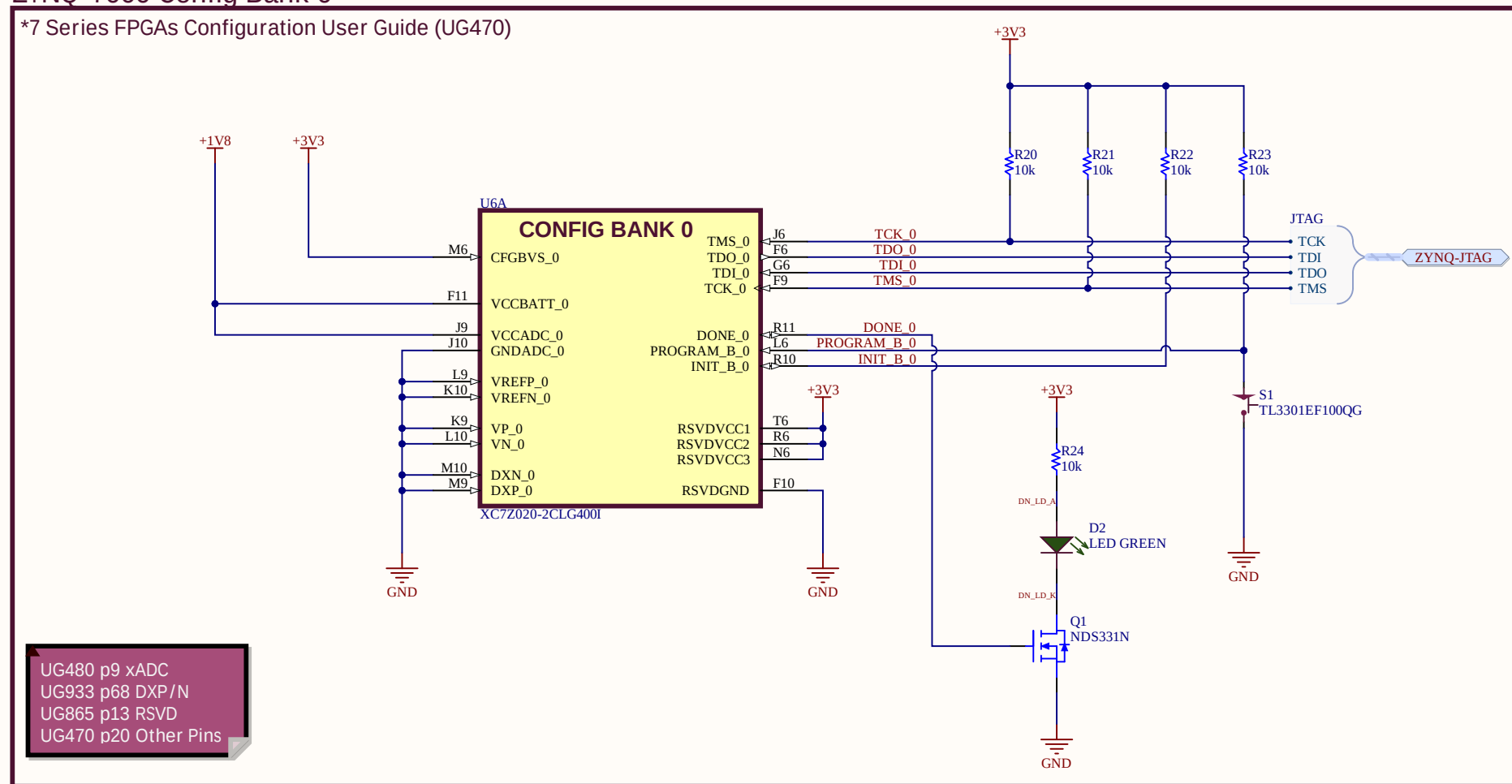


1	2	3	4	5	6	7	8
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ZHSD ZYNQ CONFIG

ZYNQ-7000 Config Bank 0

*7 Series FPGAs Configuration User Guide (UG470)



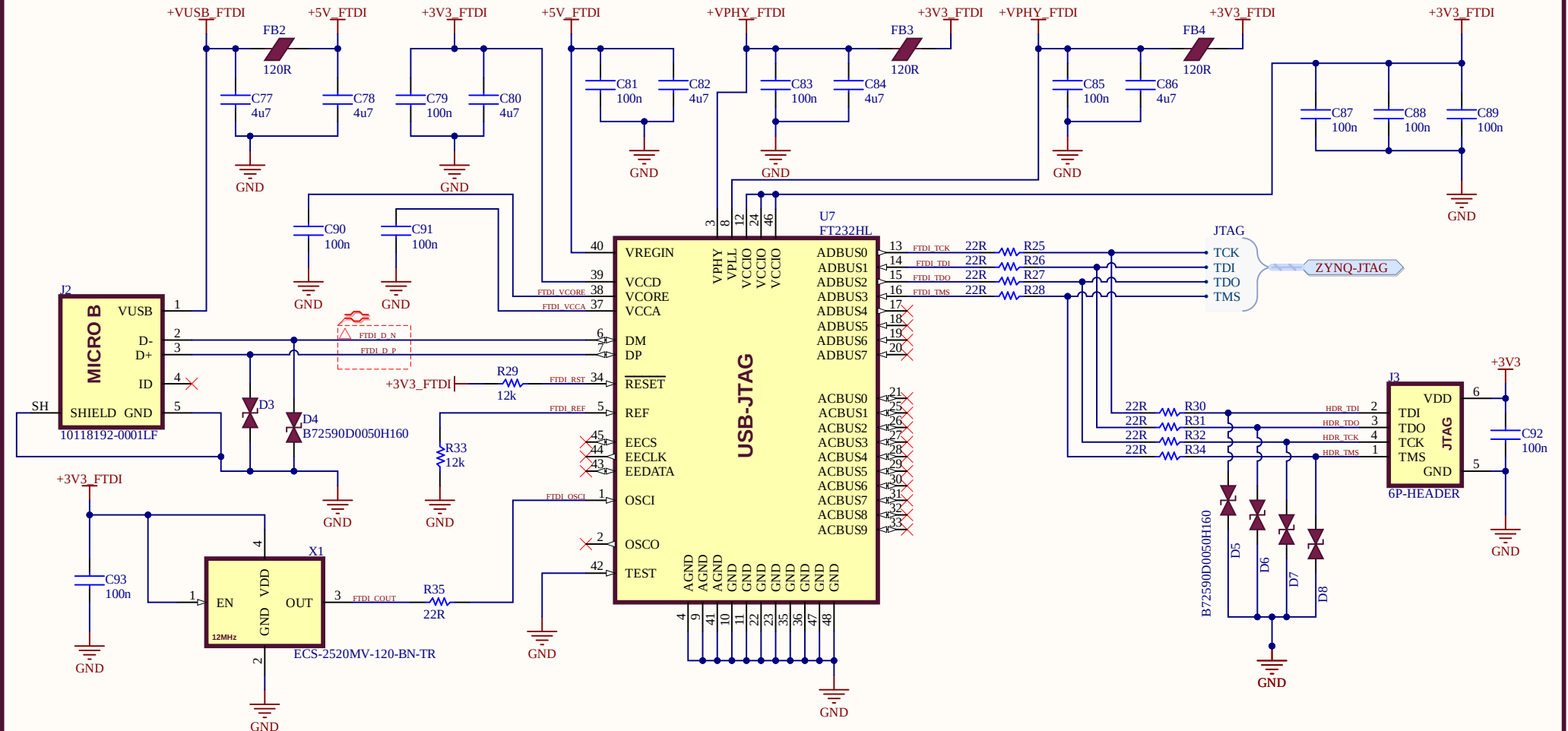
UG480 p9 xADC
UG933 p68 DXP/N
UG865 p13 RSVD
UG470 p20 Other Pins

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USB-JTAG INTERFACE

USB-JTAG

*Programming and Debugging UG908 -> Appendix E for devices

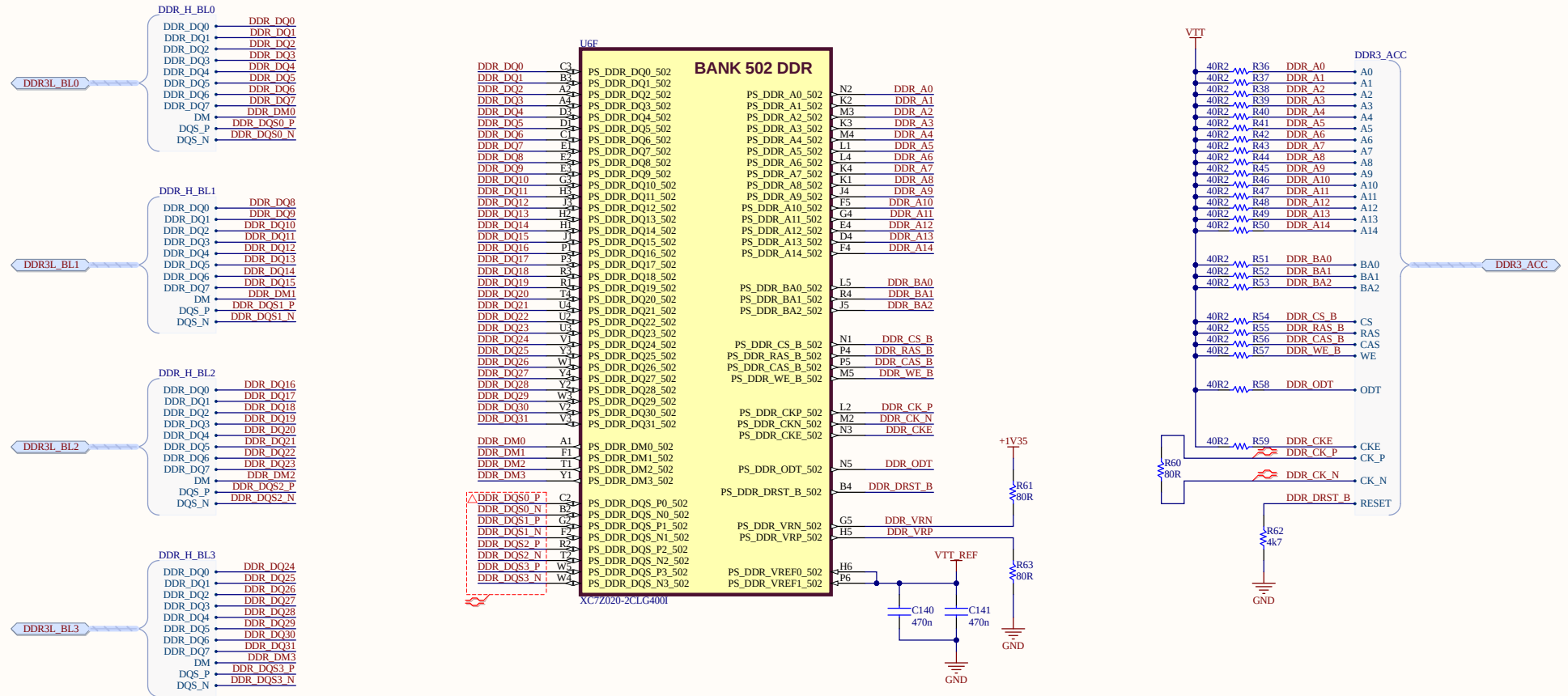


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ZHSD ZYNQ DDR3L

ZYNQ-7000 DDR BANK 502

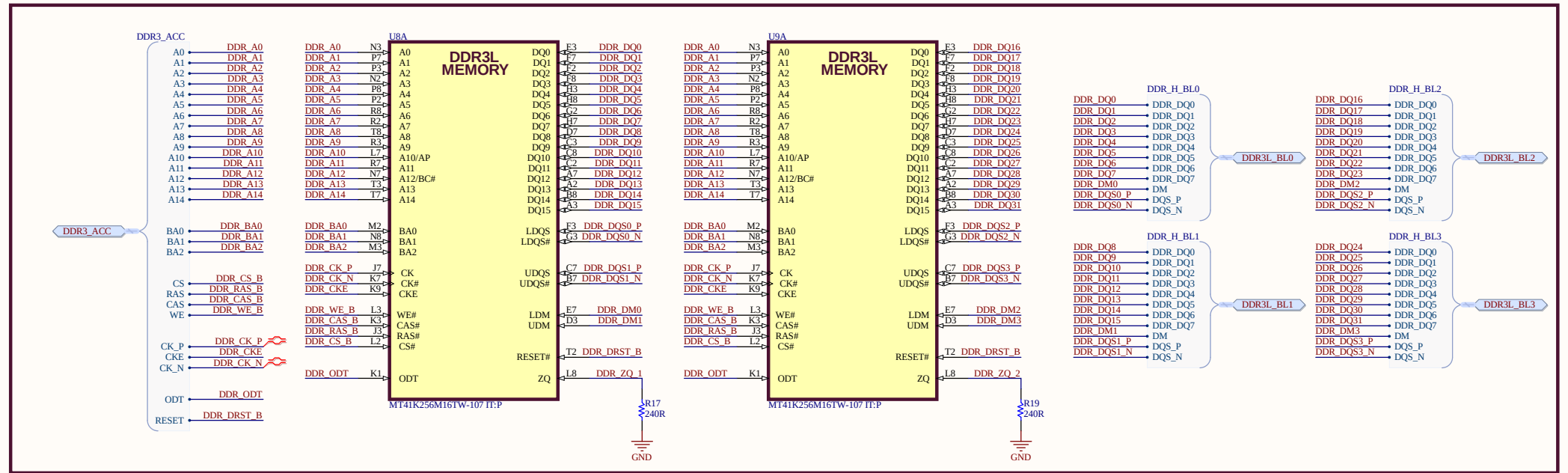
*Zynq-7000 SoC PCB Design Guide UG933



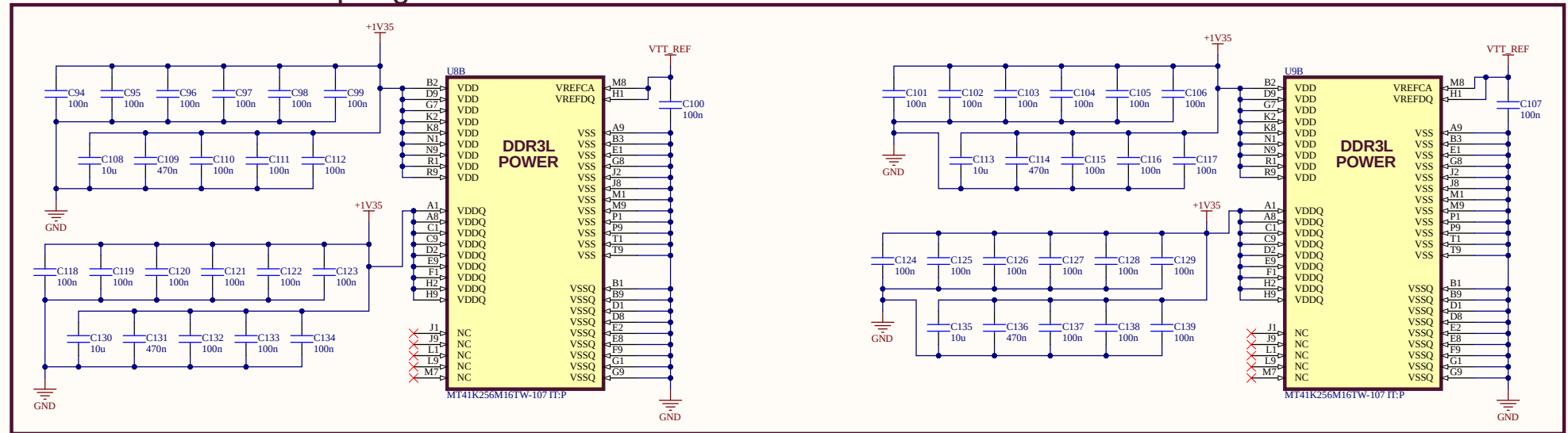
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Date:	1.30.2026	Sheet of
File:	C:\Users\ZHSD-ZYNQ-DDR\SchDoc	Drawn By:

ZHSD DDR3L 1GB

DDR3L IC



DDR3 IC Power Decoupling



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File:	C:\Users\ZHSD\Documents\ZHSD-DDR3L.SchDoc	Drawn By:

A

B

C



OTG

- OTG_DATA0 → DATA0
- OTG_DATA1 → DATA1
- OTG_DATA2 → DATA2
- OTG_DATA3 → DATA3
- OTG_DATA4 → DATA4
- OTG_DATA5 → DATA5
- OTG_DATA6 → DATA6
- OTG_DATA7 → DATA7

OTG_USB

PS

- PS_QSPI_I00_M3 → QSPI_I01
- PS_QSPI_I01_M1 → QSPI_I02
- PS_QSPI_I02_M2 → QSPI_I03
- PS_QSPI_I03_M0 → QSPI_I04
- PS_QSPI_SCK_M4 → QSPI_SCK
- PS_QSPI_SS → QSPI_SS

RS485-CAN

- PS_RS485_TX → RS485_TX
- PS_RS485_RX → RS485_RX
- PS_RS485_DE → RS485_DE
- PS_CAN_TX → CAN_TX
- PS_CAN_RX → CAN_RX
- PS_CAN_STB → CAN_STB

The diagram shows a 3V3 voltage regulator circuit. A 3V3_PG signal source is connected to a 20k resistor (R74) and a 100nF capacitor (C150) to ground. The other end of R74 is connected to the input of the U13 NL17SZ17DET2G Schmitt trigger (pin 2). The input of the Schmitt trigger is also connected to a 3V3_PG signal. The output of the Schmitt trigger (pin 4) is connected to the PS_POR signal. The Schmitt trigger is powered by a 3V3 supply (pin 1) and has its ground (pin 3) connected to ground. A 100nF capacitor (C149) is connected between the 3V3 supply and ground. A 20k resistor (R80) is connected between the output of the Schmitt trigger and ground.

[illegible]

The diagram illustrates the eMMC interface circuit. The eMMC controller is connected to the eMMC device. The controller's pins are labeled as follows:

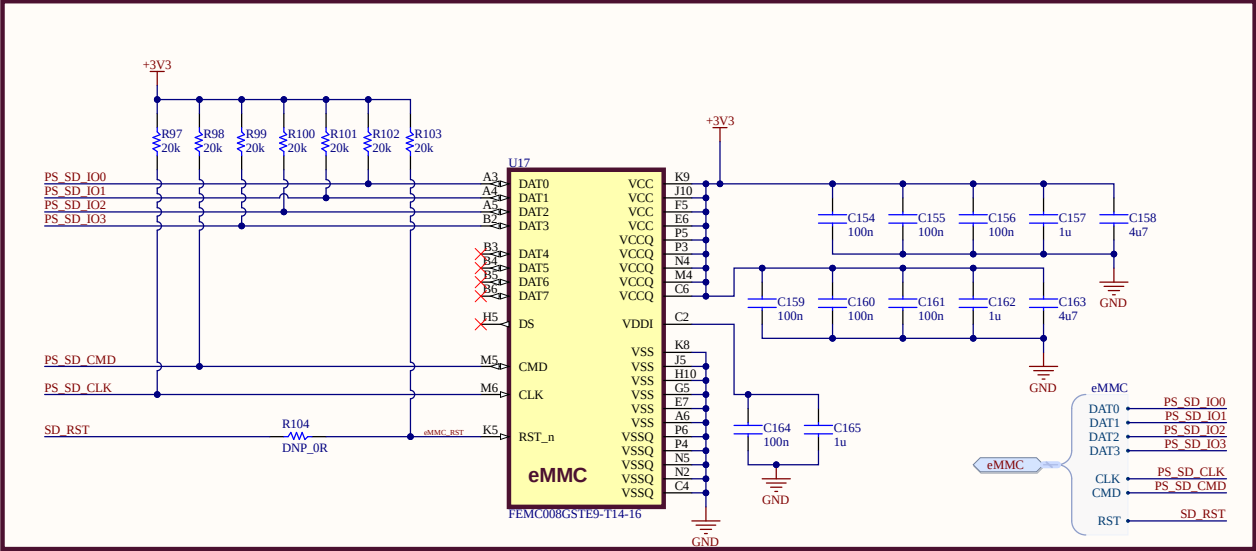
- DAT0, DAT1, DAT2, DAT3: PS_SD IO0, PS_SD IO1, PS_SD IO2, PS_SD IO3
- CLK: PS_SD CLK
- CMD: PS_SD CMD
- RST: SD_RST

The RST pin is connected to a +1V8 supply through a 20k resistor (R83). A MOSFET (Q3, BSS138LT1G) is connected between the RST pin and ground.

1	2	3	4	5	6	7	8
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ZHSD BOOTING & STORAGE

eMMC Storage



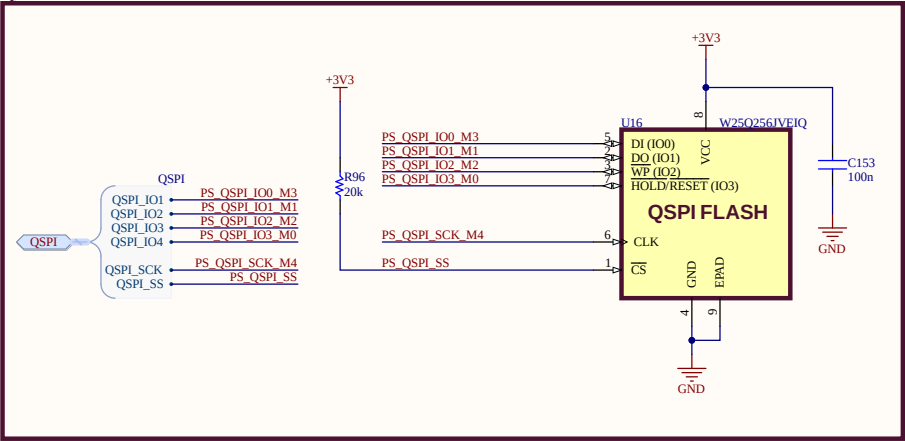
ZYNQ Boot Modes Reference

Table 6-4: Boot Mode MIO Strapping Pins

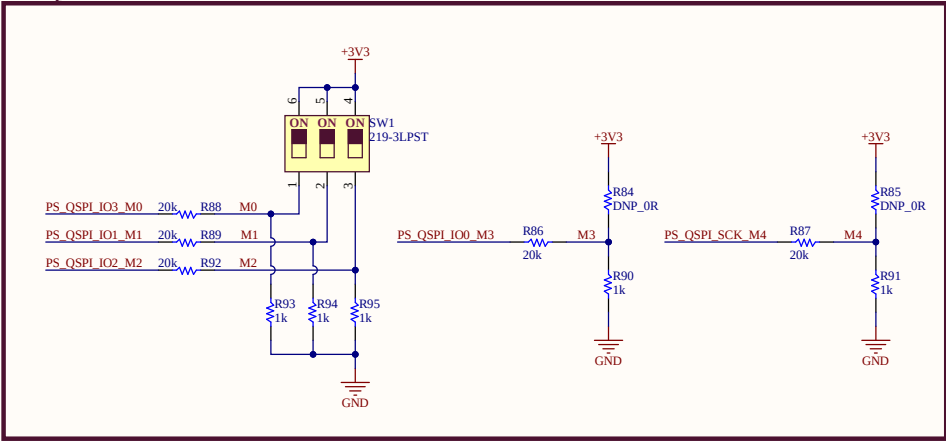
Pin-signal / Mode	MIO[8]	MIO[7]	MIO[6]	MIO[5]	MIO[4]	MIO[3]	MIO[2]
	VMODE[1]	VMODE[0]	BOOT_MODE[4]	BOOT_MODE[0]	BOOT_MODE[2]	BOOT_MODE[1]	BOOT_MODE[3]
Boot Devices							
JTAG Boot Mode; cascaded is most common ⁽¹⁾		0	0	0		JTAG Chain Routing ⁽²⁾ 0: Cascade mode 1: Independent mode	
NOR Boot ⁽³⁾		0	0	1			
NAND		0	1	0			
Quad-SPI ⁽³⁾		1	0	0			
SD Card		1	1	0			
Mode for all 3 PLLs							
PLL Enabled		0	Hardware waits for PLL to lock, then executes BootROM.				
PLL Bypassed		1	Allows for a wide PS_CLK frequency range.				
MIO Bank Voltage ⁽⁴⁾							
	Bank 1	Bank 0	Voltage Bank 0 includes MIO pins 0 thru 15.				
2.5 V, 3.3 V	0	0	Voltage Bank 1 includes MIO pins 16 thru 53.				
1.8 V	1	1					

- Notes:
1. JTAG cascaded mode is most common and is the assumed mode in all the references to JTAG mode except where noted.
 2. For secure mode, JTAG is not enabled and MIO[2] is ignored.
 3. The Quad-SPI and NOR boot modes support execute-in-place (this support is always non-secure)
 4. Voltage Banks 0 and 1 must be set to the same value when an interface spans across these voltage banks. Examples include NOR, 16-bit NAND, and a wide TPIU test port. Other interface configuration may also span the two banks.

QSPI Flash

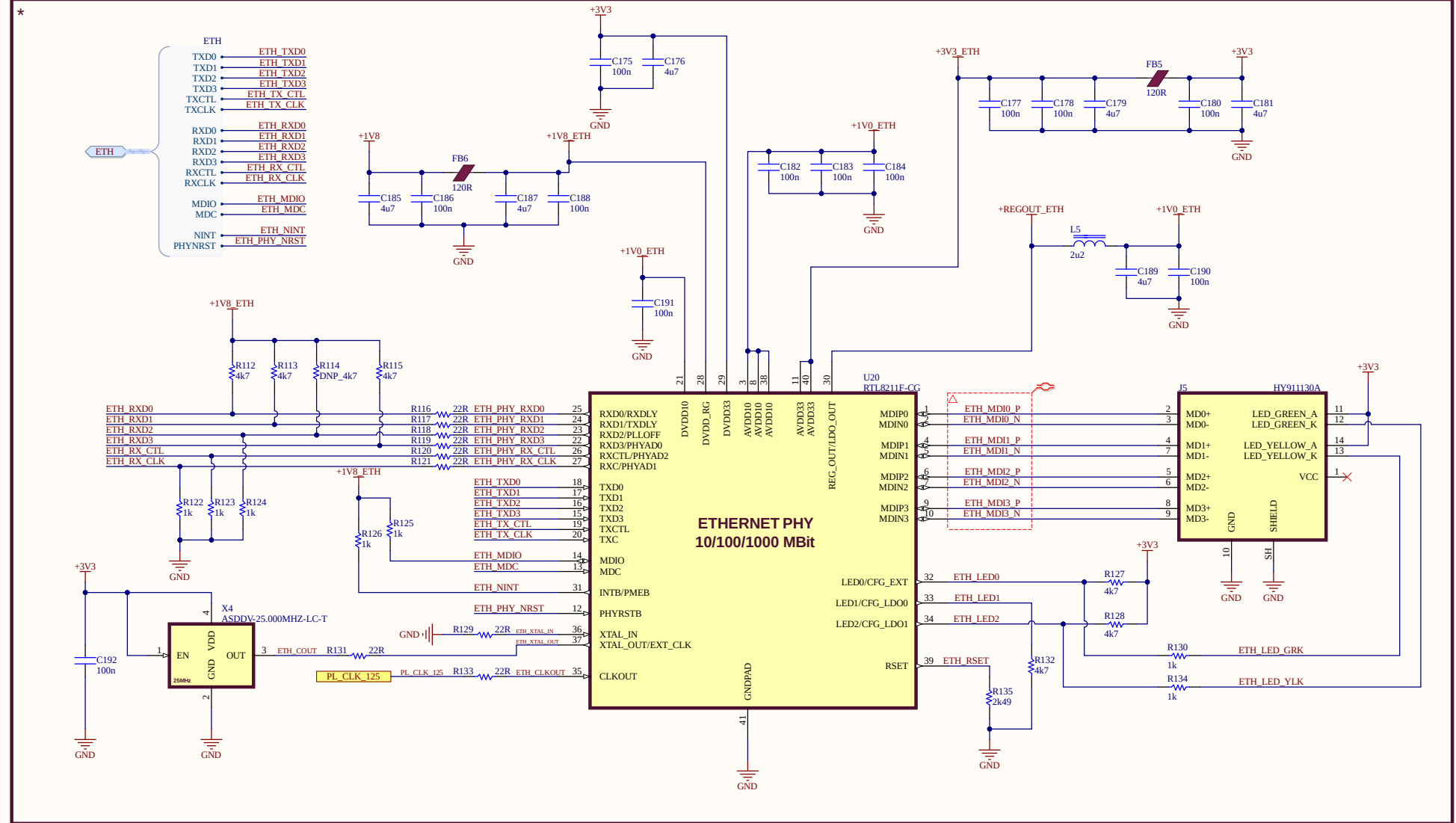


ZYNQ Boot Modes



ZHSD GIGABIT ETHERNET

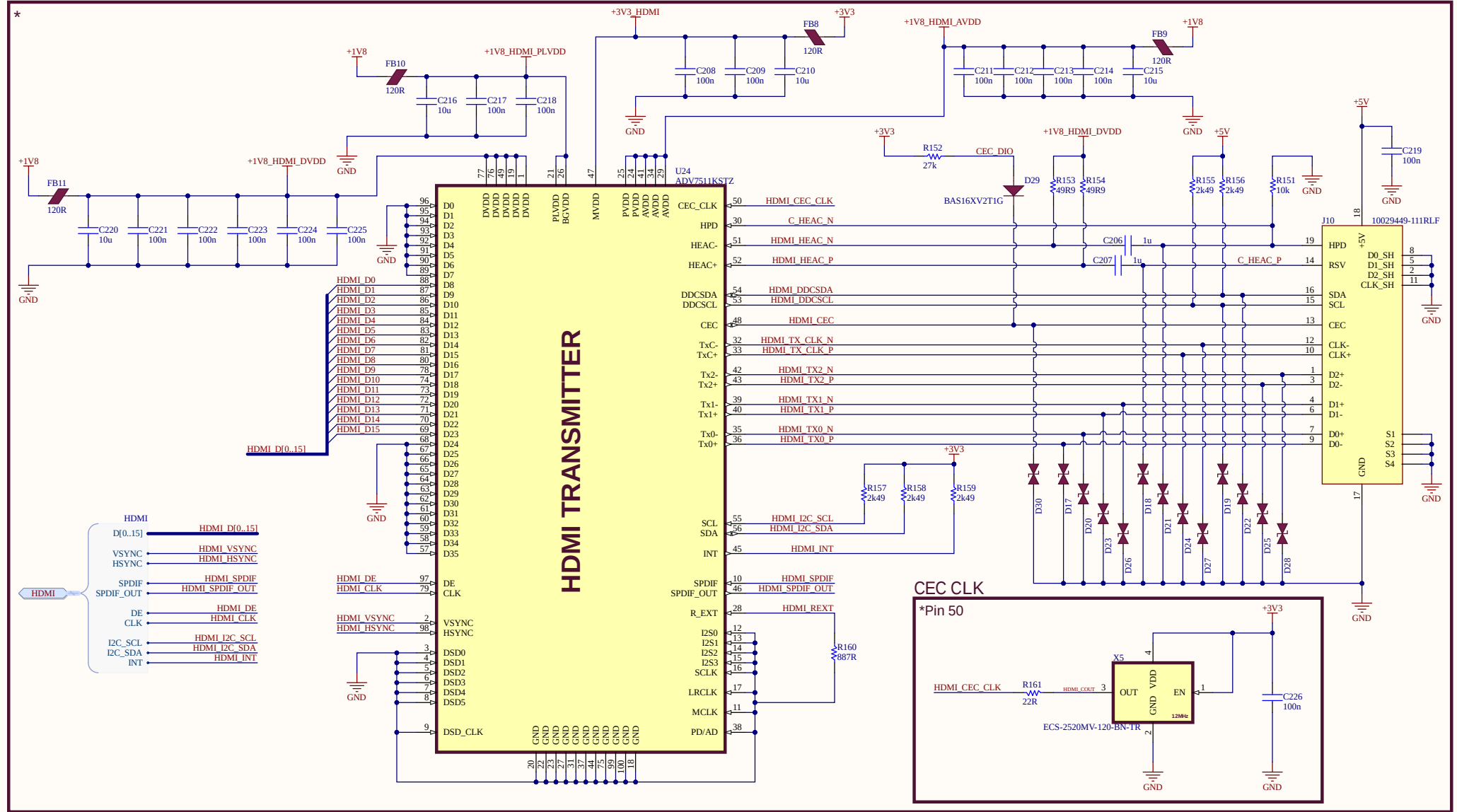
Ethernet PHY & Magjack



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ZHSD HDMI INTERFACE

HDMI Transmitter and Connector

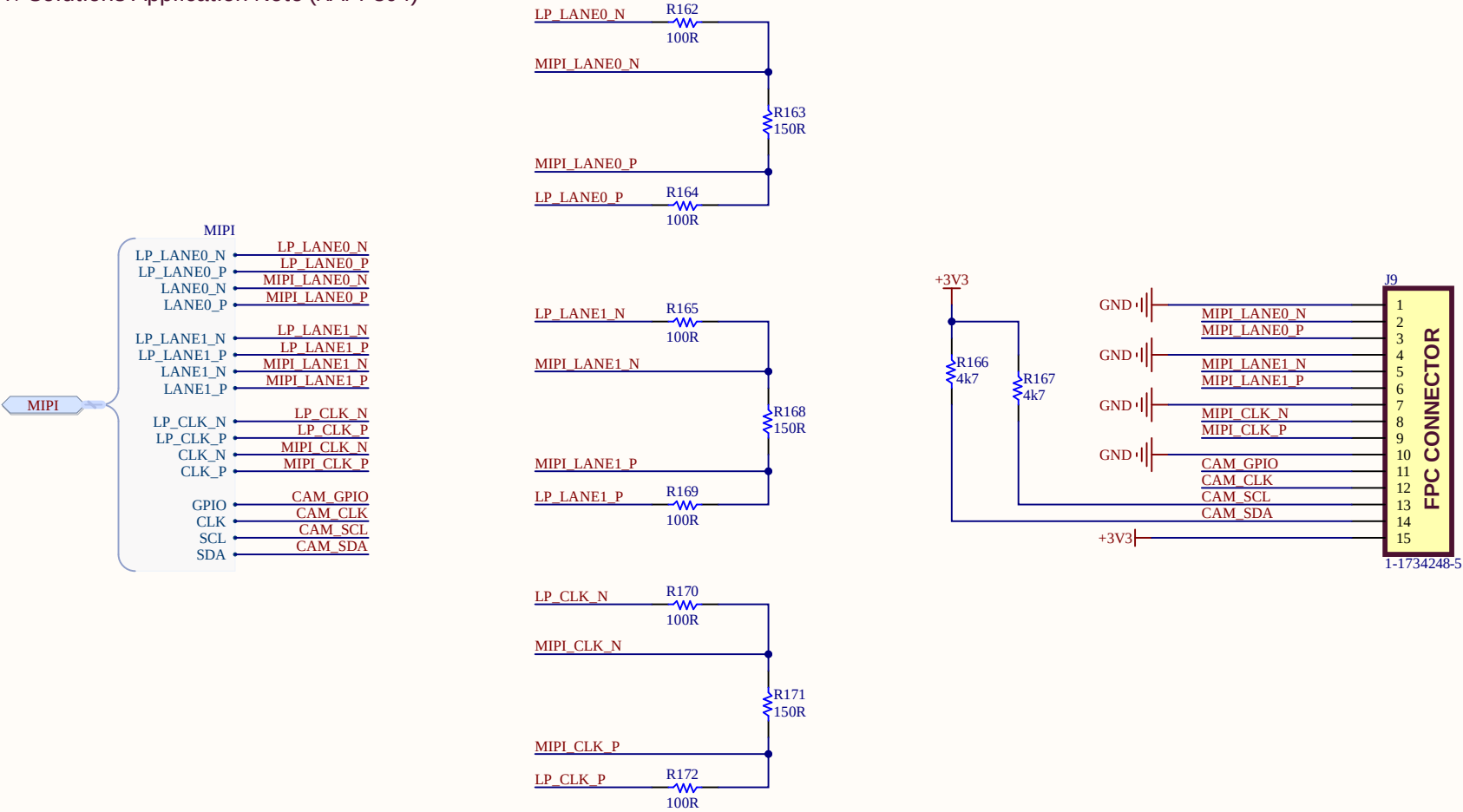


Title		
Size	Number	Revision
A3		
Date:	1.30.2026	Sheet of
File:	C:\Users\ZHSD\HDMI\SchDoc	Drawn By:

ZHSD MIPI INTERFACE

MIPI CSI-2

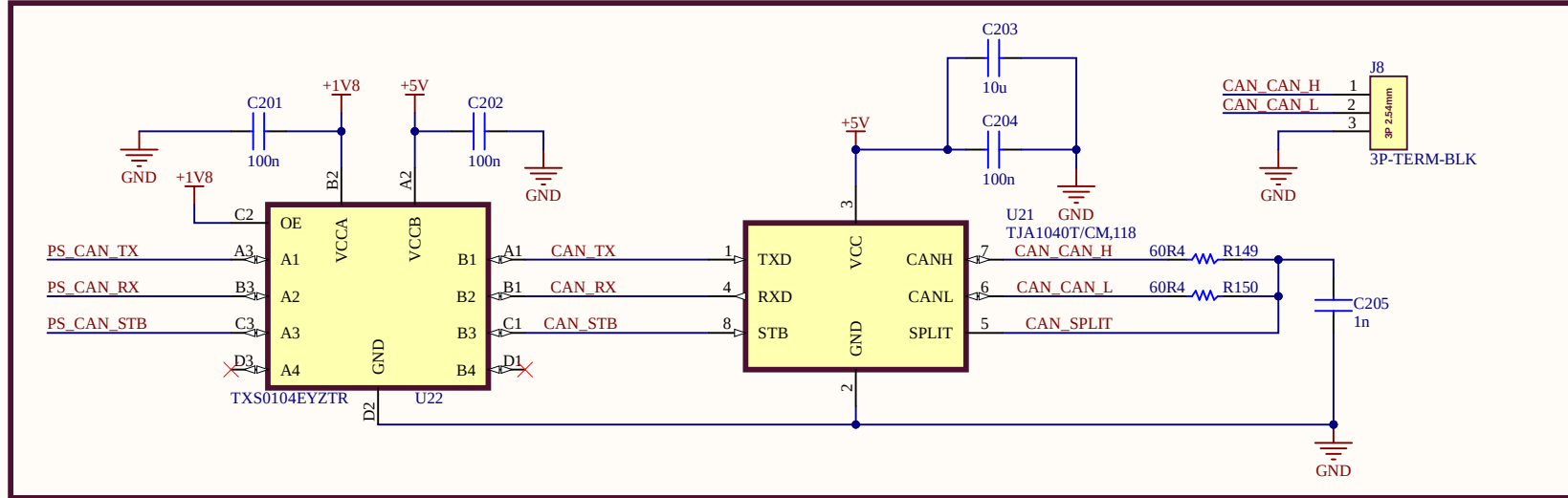
*D-PHY Solutions Application Note (XAPP894)



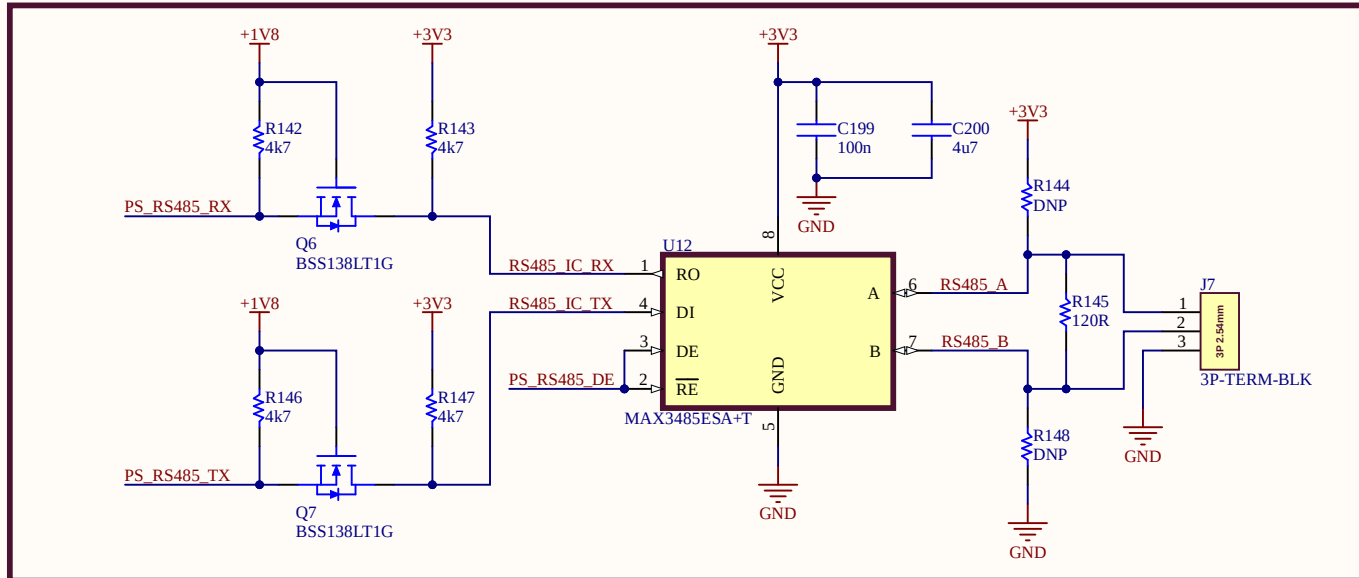
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RS485 & CAN INTERFACES

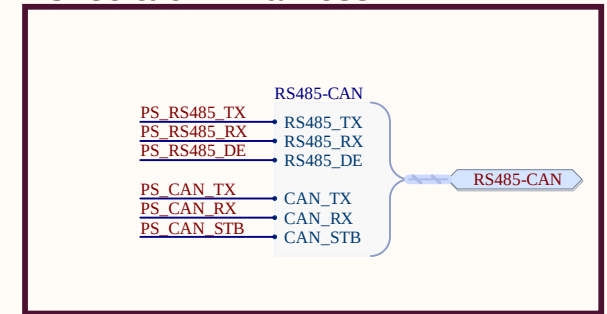
CAN Transceiver & Level Translation



RS485 Transceiver



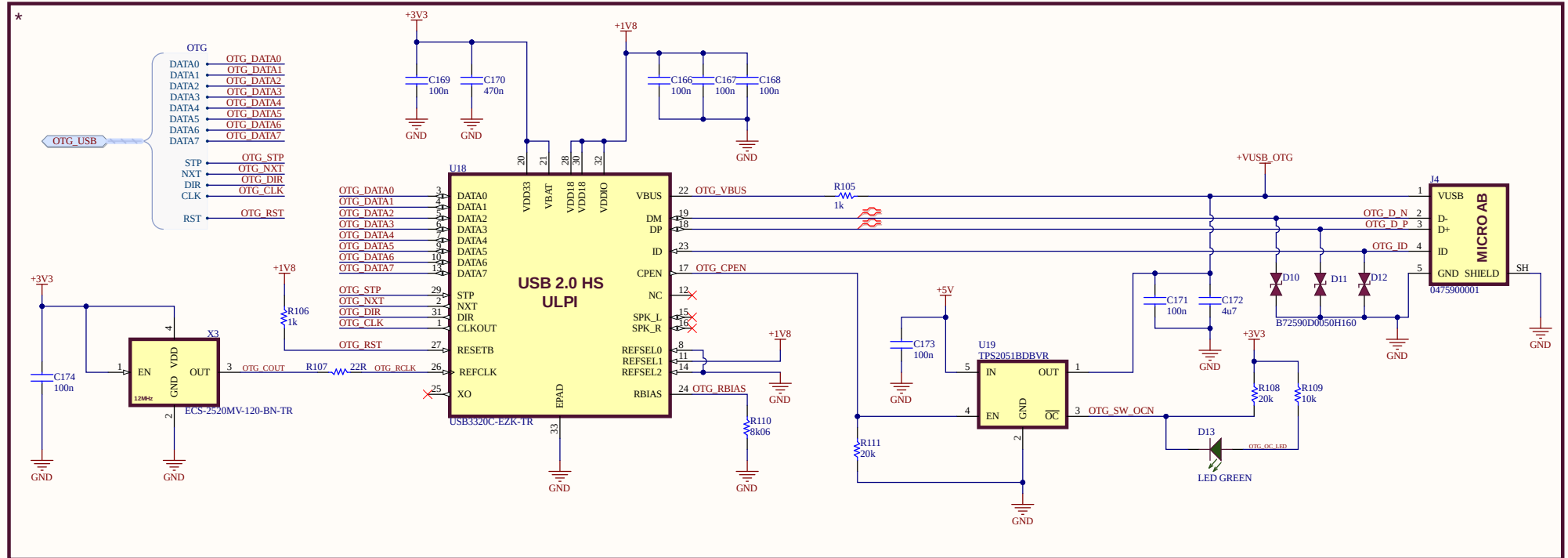
RS485 & CAN Harness



Title		
Size	Number	Revision
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ZHSD USB OTG 2.0 ULPI

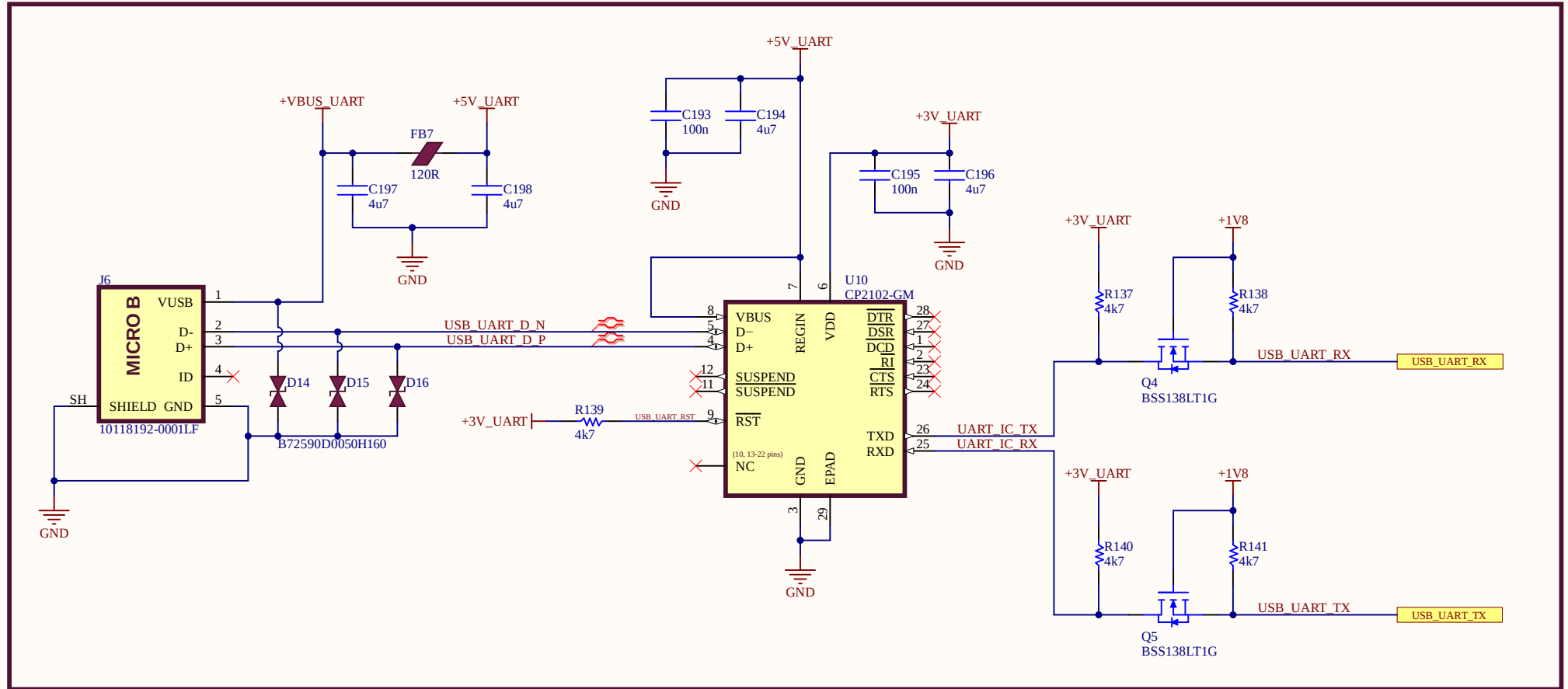
USB OTG PHY & Micro AB Connector



Title		
Size A3	Number	Revision
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USB-UART INTERFACE

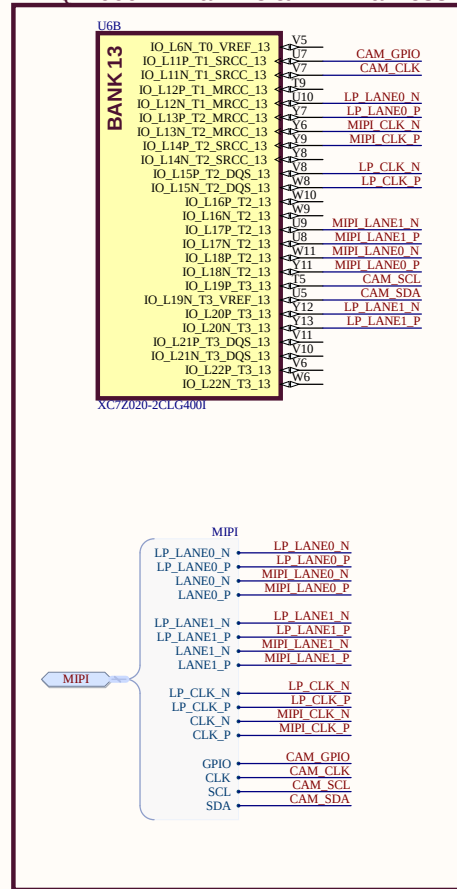
USB-UART



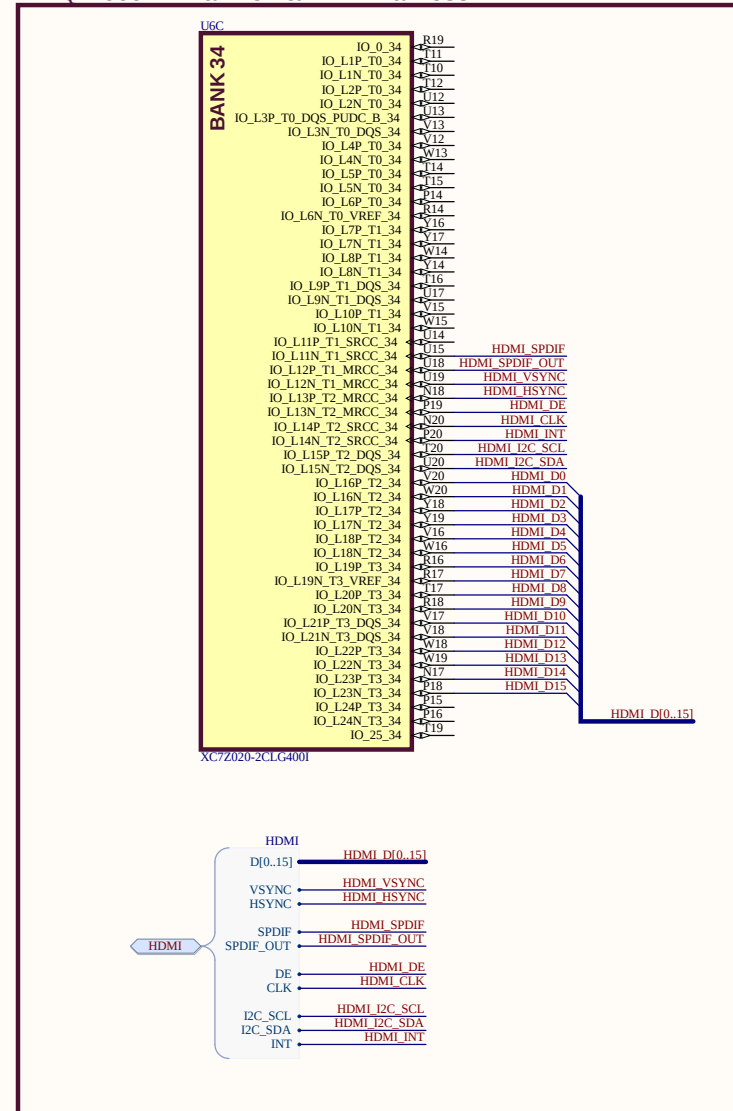
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ZHSD ZYNQ PL B ANKS

ZYNQ-7000 PL Bank 13 & MIPI Harness



ZYNQ-7000 PL Bank 34 & HDMI Harness



ZYNQ-7000 PL Bank 35

